

Integrated mechanics discovery summary: four regime-enlarged computational claims

7T10 structure-contact tensile mechanics

A. 7T10 structure-contact tensile mechanics

Accepted: linear force-extension model
Rejected: mean-force null model
Gate: AIC; improvement=69.50

contact_entropy_nats: 1.78
contact_gini: 0.0946
hotspot_load_anchor_index: 1

Claim: 7T10 supports a contact-localized tensile mechanics interpretation: a small hotspot set concentrates structural anchoring while the force trace passes a linear tensile-response gate with measurable stiffness, work, and peak force.

Fiber-network anisotropic mechanics

B. Fiber-network anisotropic mechanics

Accepted: orientation-tensor anisotropic stiffness surrogate
Rejected: isotropic fiber-count descriptor
Gate: AIC; improvement=123.87

anisotropy_ratio: 5.12
orientation_order_parameter: 0.673
stiffness_kpa: 119

Claim: The fiber-network run supports an anisotropic mechanics claim: orientation eigenstructure defines a dominant load-bearing axis while the stress-strain surrogate supplies the tensile stiffness scale.

Mechanobiology force-path mechanics

C. Mechanobiology force-path mechanics

Accepted: full force-path regression
Rejected: adhesion-only traction model
Gate: BIC; improvement=368.10

load_concentration: 0.103
mean_load_path_score_pa_per_um: 4.42
strongest_path_id: 12

Claim: The mechanobiology run supports a graph-mediated load-routing claim: traction is best explained as a force-path property combining adhesion, cytoskeletal coupling, displacement, and path length, not as adhesion alone.

Membrane curvature-energy mechanics

D. Membrane curvature-energy mechanics

Accepted: Helfrich-style curvature-energy proxy
Rejected: curvature-only shape descriptor
Gate: criterion; improvement=0.00

curvature_energy_localization_top10_fraction: 0.327
max_abs_curvature_1_um: 0.295
mean_energy_density_proxy_kbt_per_um2: 0.239

Claim: The membrane run supports a curvature-energy mechanics claim: the synthetic curvature field has localized bending-energy structure whose total magnitude is controlled by the assigned bending modulus.